Multi-Agent Education System

Course Name: AGENTIC AI

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

Sr no	Student Name	Enrolment Number
	SUJAL MUKATI	EN22CS301995
	SUMAN DANGI	EN22CS301998
	TANISHQ SHARMA	EN22CS3011021
	UDHHAV CHOURASIYA	EN22CS3011042
	VAISHNAVI SARODEY	EN22CS3011056

Group Name:04D9

Project Number:03

Industry Mentor Name: Mr. Ashwin Sir

University Mentor Name: Prof. Pradeep Baniya

Academic Year:2025-2026

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features
2. Tools & Technologies Used
3. Overall Architecture / Workflow

Results & Output

1. *Screenshots / outputs*
2. *Key outcomes*

Conclusion

Future Scope & Enhancements

Problem Statement & Objectives

1. Problem Statement

In today's highly competitive job market, interview preparation has become a critical factor for career success. However, many students and job seekers face significant challenges in preparing effectively. One of the major issues is the lack of structured and personalized learning resources. Most candidates rely on scattered online materials, which are often outdated, generic, and not tailored to specific job roles.

Another key challenge is the absence of real-time evaluation and feedback. Traditional preparation methods such as reading notes or practicing questions do not provide insights into the correctness, clarity, or quality of answers. As a result, candidates are unable to identify their weaknesses or improve their performance effectively.

Additionally, mock interviews conducted by humans require time, availability, and resources, making them inaccessible to many learners. There is also inconsistency in feedback quality, which can affect the learning process.

With the rapid advancement of Artificial Intelligence, there is an opportunity to automate and enhance interview preparation. However, most existing systems are limited to single-task solutions, such as only generating questions or only evaluating answers, without providing a complete end-to-end solution.

Therefore, there is a need for an intelligent, scalable, and automated system that can:

- Generate role-specific interview questions
- Evaluate user responses accurately
- Provide personalized feedback and suggestions
- Simulate a real interview environment
- Deliver structured performance reports

To address these challenges, this project proposes a **Multi-Agent Education System**, where multiple AI agents collaborate to perform different tasks efficiently. This approach ensures better accuracy, modularity, and a more realistic and effective interview preparation experience.

2. Project Objectives

- To design a smart interview preparation system using AI
- To simulate real-world interview scenarios
- To generate role-based and dynamic questions

- To evaluate answers intelligently using LLMs
- To provide personalized feedback and suggestions
- To generate structured performance reports
- To improve user confidence and learning outcomes

3. Scope of the Project

The scope of this project defines the range of functionalities and the potential applications of the Multi-Agent Education System. This system is designed to assist users in interview preparation by automating question generation, answer evaluation, and feedback using Artificial Intelligence.

The project primarily focuses on supporting multiple job roles such as Software Developer, Data Analyst, and other technical domains. It can generate role-specific questions and evaluate descriptive answers, making it suitable for both beginners and advanced learners.

The system is built using a scalable architecture with tools like CrewAI for multi-agent collaboration and Gemini 3.1 Flash API for intelligent processing. This allows the system to be easily extended by adding new agents, features, or domains without major changes to the existing structure.

Currently, the system provides:

- Question generation
- Answer evaluation
- Personalized feedback
- Report generation (PDF)
- Interactive interface using Streamlit

The project is mainly limited to text-based interaction and does not include advanced features like voice interviews or real-time video analysis. However, it provides a strong foundation for future enhancements.

Out of Scope

- Real-time video interview simulation
- Voice-based interaction or speech recognition
- Integration with external job portals (LinkedIn, Naukri, etc.)
- Resume parsing and resume-based question generation
- Live human interviewer interaction
- Advanced UI/UX design like full web applications
- Mobile application development (Android/iOS)

Limitations

- Depends on AI API accuracy (may vary sometimes)
- Requires internet connection for API calls
- Limited to text-based interaction
- Basic user interface using Streamlit
- No permanent database (data not stored long-term)
- Evaluation may not be 100% accurate like human experts
- Limited customization for different interview styles
- Performance depends on API response time

Target Audience

- Students and freshers
- Job seekers
- Training institutes
- Educators
- Self-learners

Proposed Solution

1. Key Features

Multi-Agent Architecture

The system is built using a multi-agent approach where different AI agents perform specific tasks independently.

Role-Based Question Generation

The system generates interview questions based on the selected job role.

AI-Based Answer Evaluation

The system evaluates user answers using AI via Gemini 3.1 Flash API

Personalized Feedback System

After evaluation, the system provides feedback to the user.

Automated Report Generation

The system generates a structured report in PDF format using FPDF

2. Tools & Technologies Used

1. Python → Core Programming Language

Python is used as the main programming language for developing the system.

2. Streamlit → Frontend UI

Streamlit is used to create the user interface of the application.

3. CrewAI → Agent Orchestration

CrewAI is used to manage and coordinate multiple AI agents.

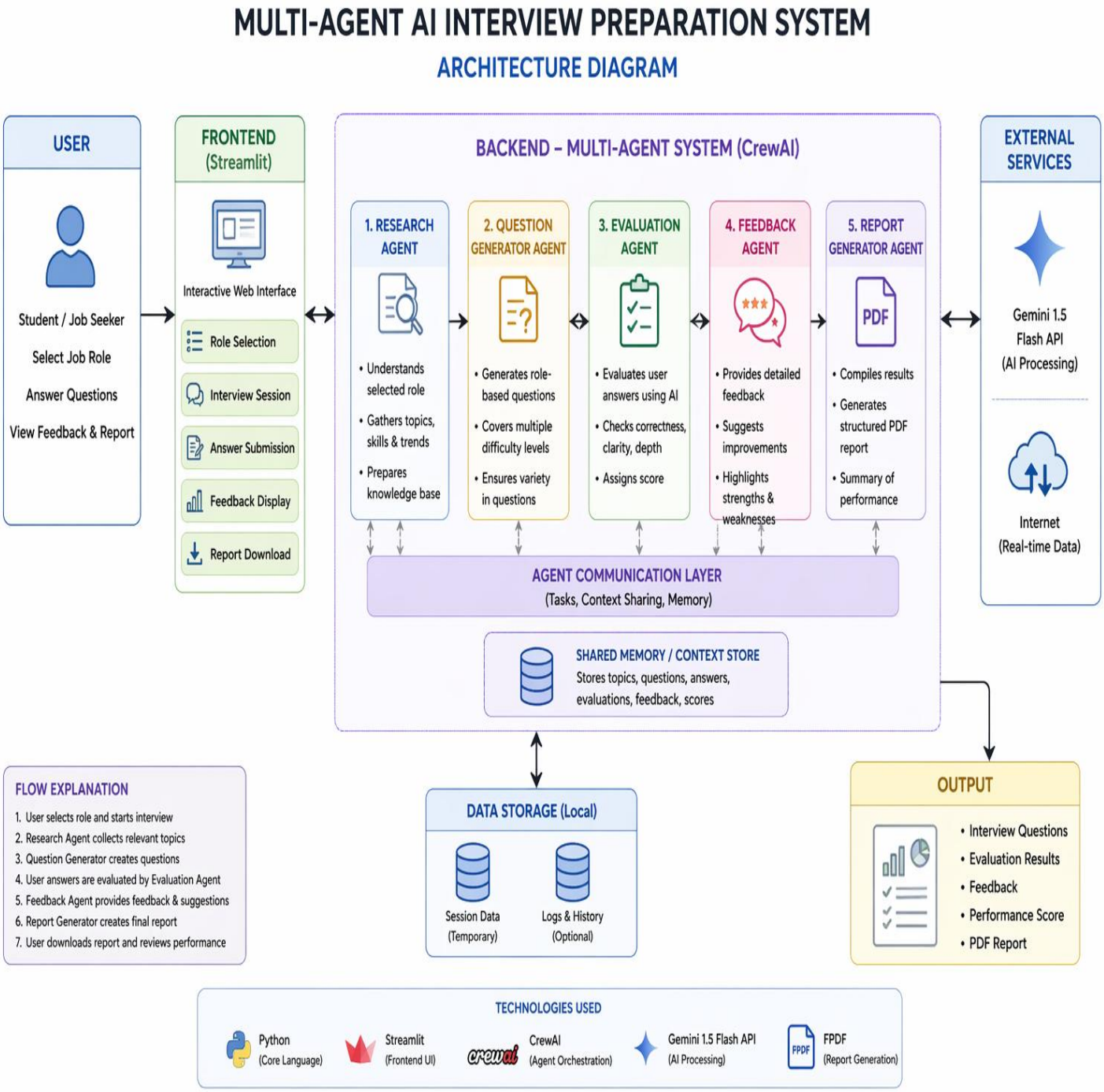
4. Google Gemini (Gemini 1.5 Flash API) → AI Processing

Gemini API is used as the Large Language Model (LLM) for processing tasks.

5. FPDF → Report Generation

FPDF is used to generate PDF reports.

3. Overall Architecture / Workflow

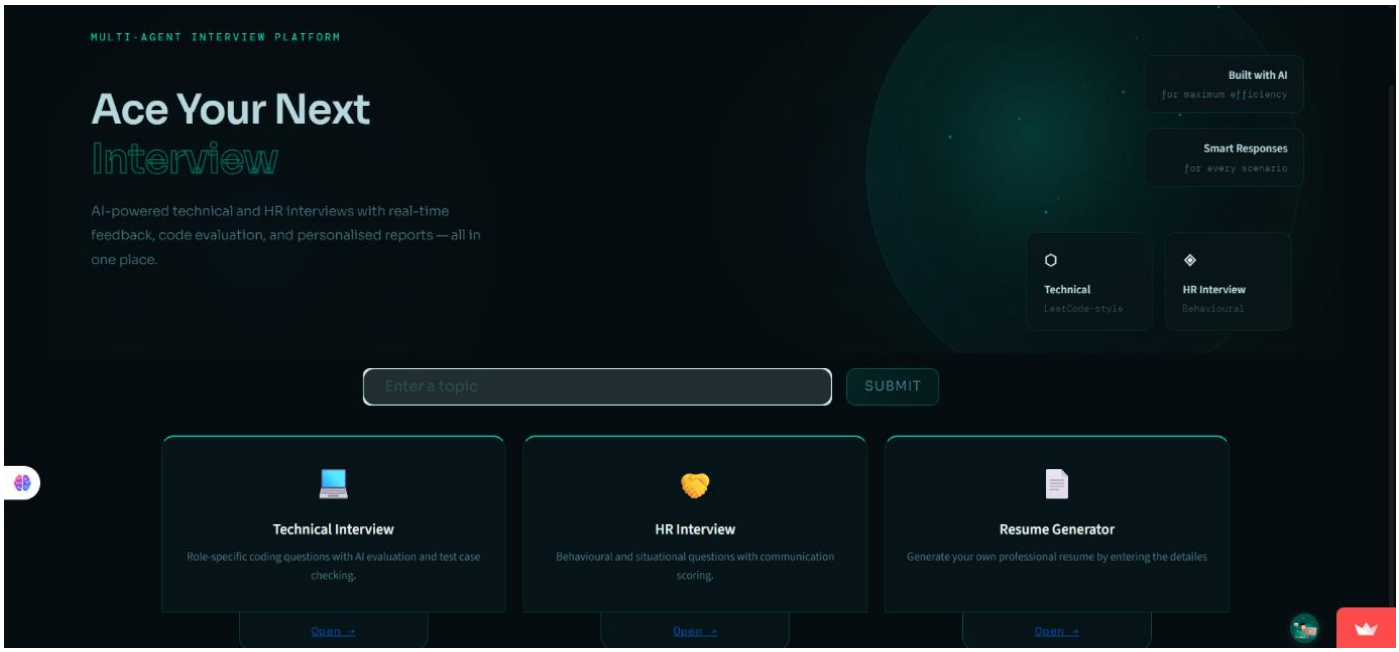


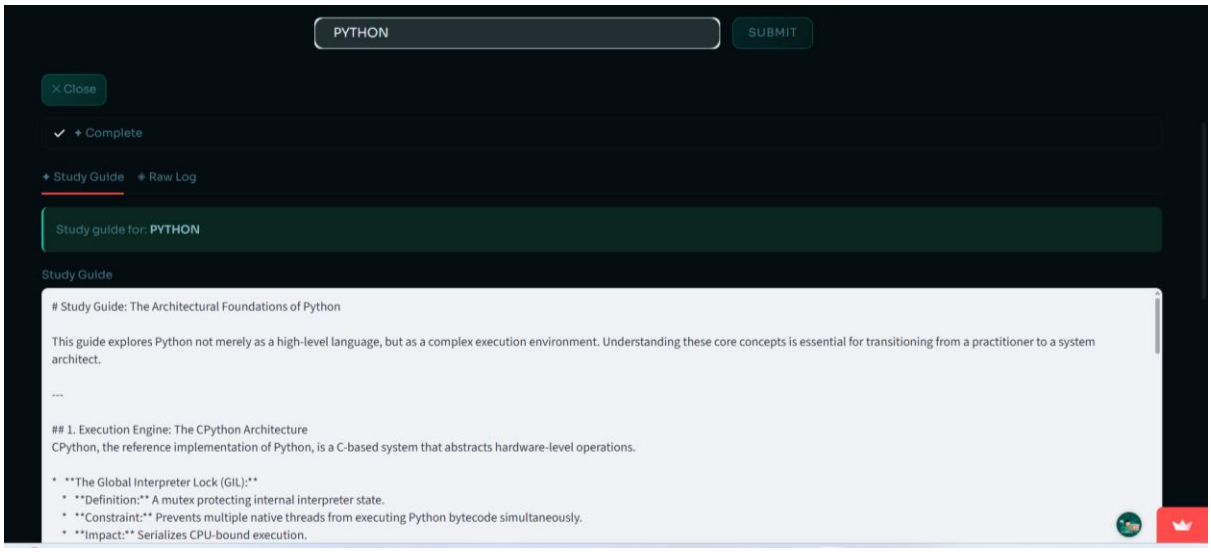
Workflow Explanation

- Define multiple agents (Research, Question Generator, Evaluation, Feedback, Report)
- Assign specific tasks to each agent
- Create a Crew (group of all agents)
- Set execution order (sequential workflow)
- Start workflow when user selects job role
- Research Agent collects topics
- Output passed to Question Generator Agent
- Questions generated and sent to user
- User submits answers
- Evaluation Agent analyzes answers
- Output passed to Feedback Agent
- Feedback Agent generates suggestions
- Output passed to Report Generator Agent
- Report Agent compiles final report
- CrewAI manages data flow and communication between all agents

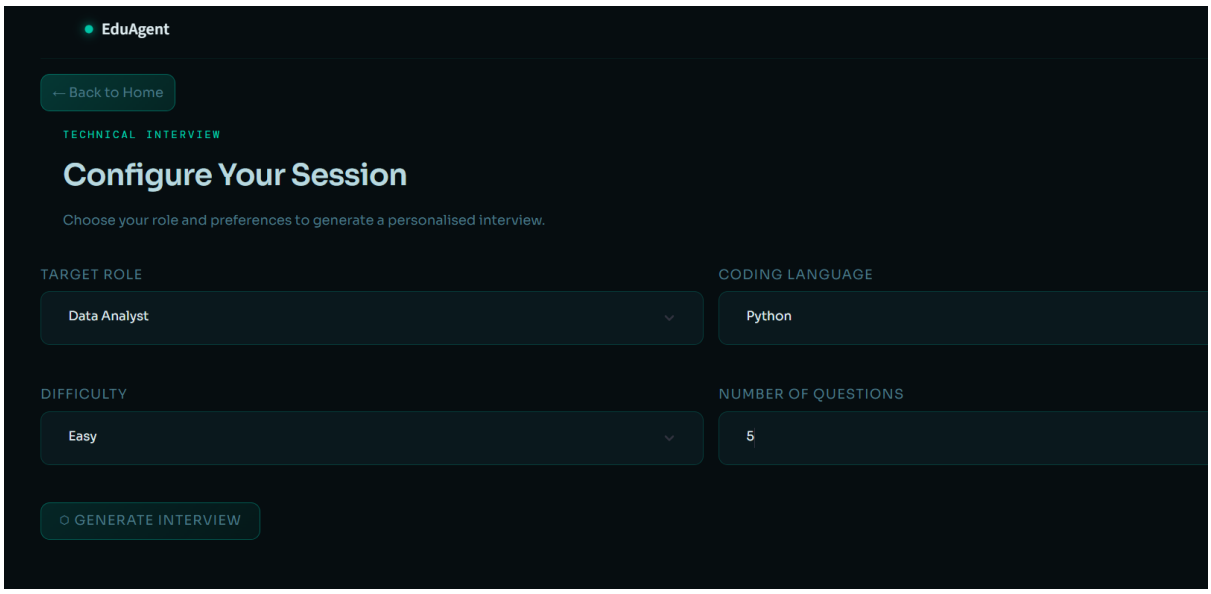
Results & Output

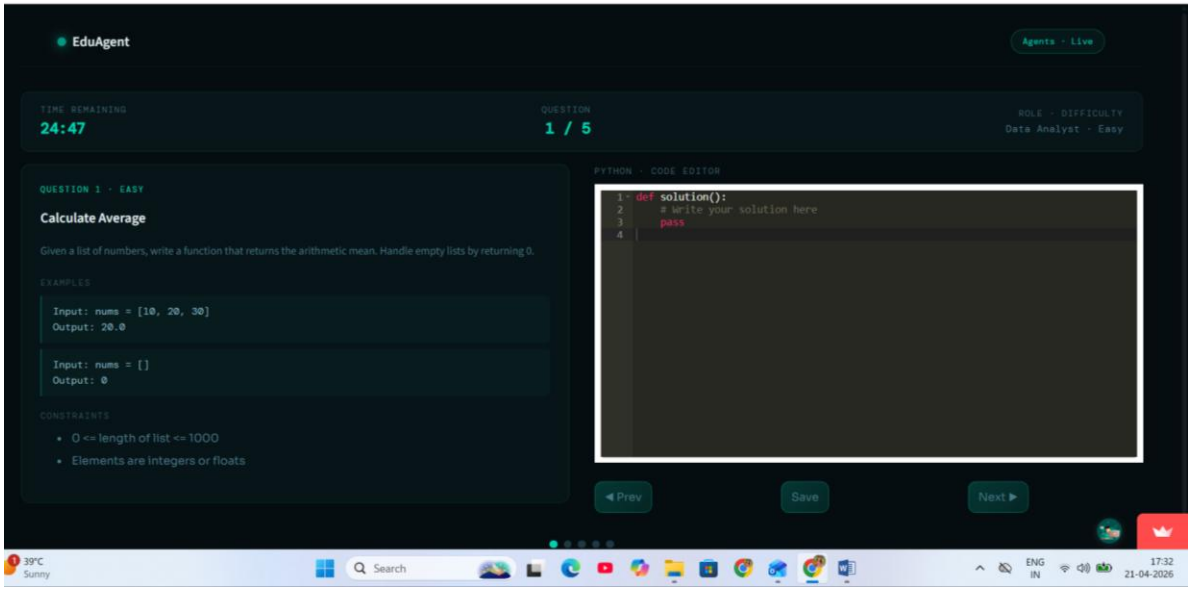
1. Screenshots / Outputs

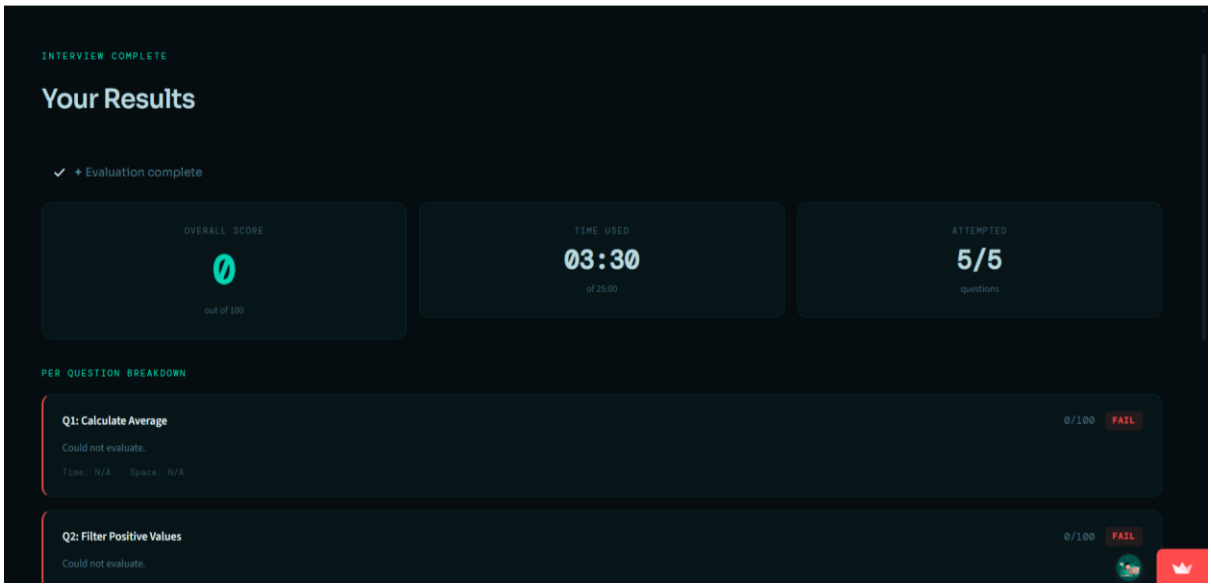




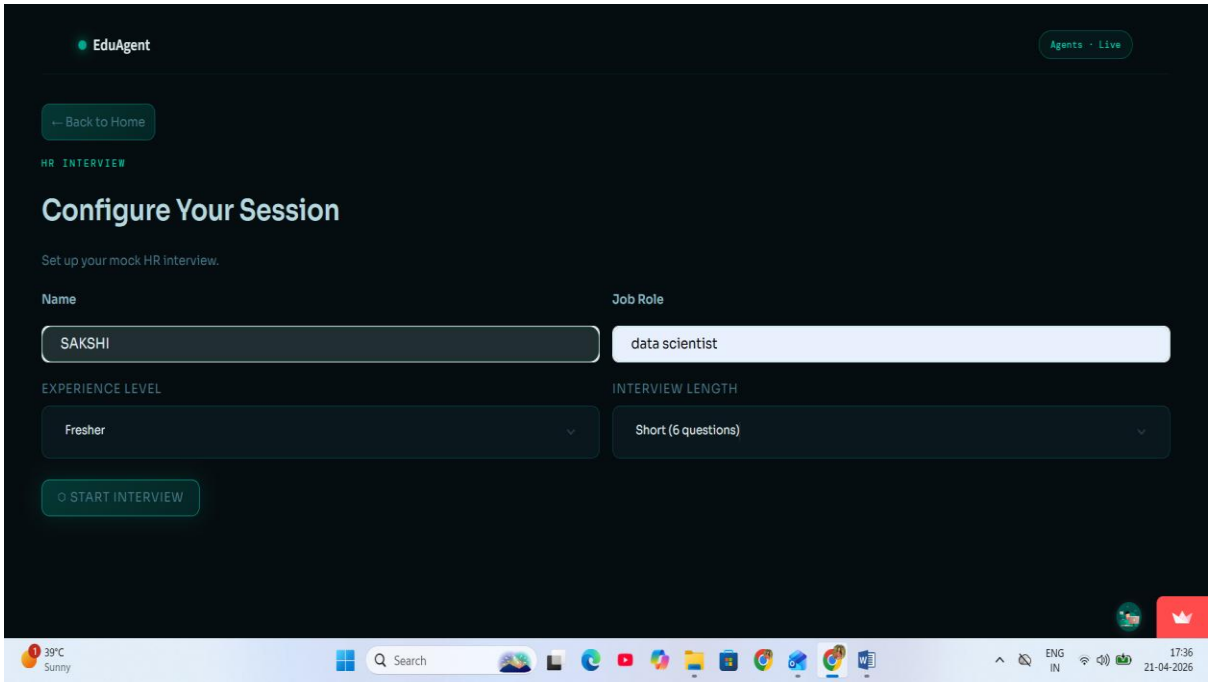
Technical Interview:

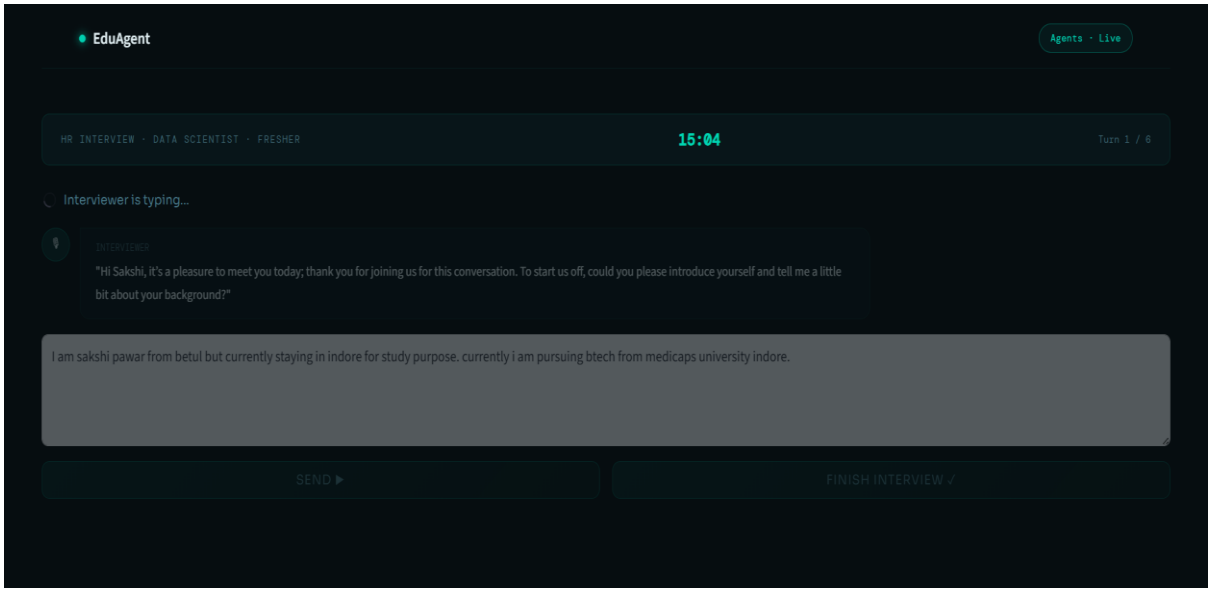


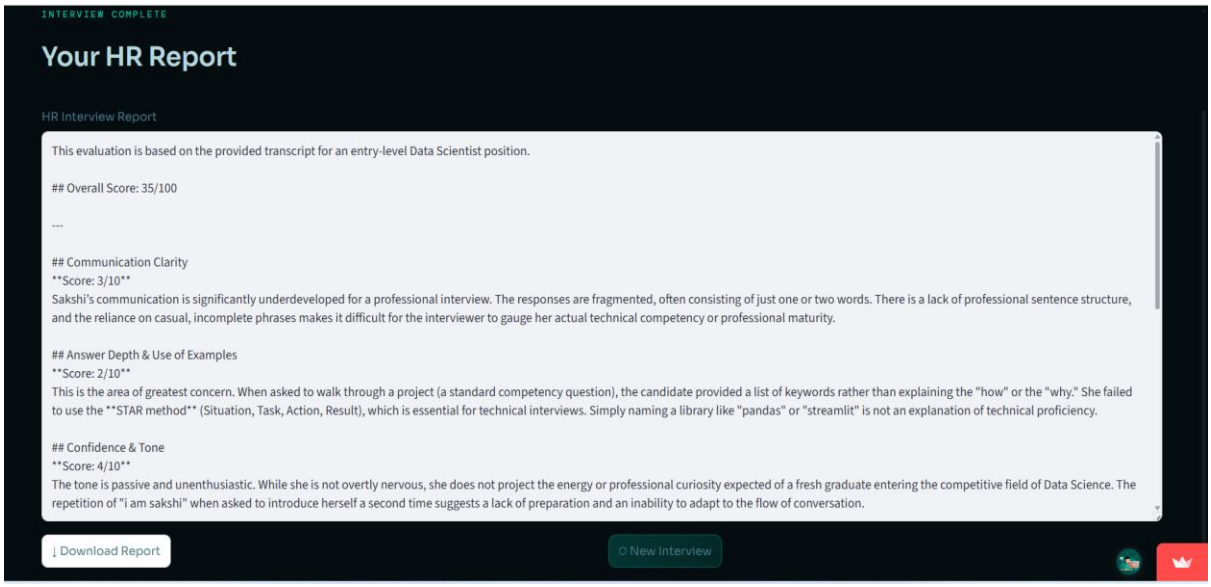

EduAgent
Agents · Live
TIME REMAINING
24:47
QUESTION
1 / 5
ROLE · DIFFICULTY
Data Analyst · Easy
QUESTION 1 · EASY
Calculate Average
Given a list of numbers, write a function that returns the arithmetic mean. Handle empty lists by returning 0.
EXAMPLES
Input: nums = [10, 20, 30]
Output: 20.0
Input: nums = []
Output: 0
CONSTRAINTS
0 <= length of list <= 1000
Elements are integers or floats
PYTHON · CODE EDITOR
def solution():
# write your solution here
pass
Prev
Save
Next


INTERVIEW COMPLETE
Your Results
Evaluation complete
OVERALL SCORE
0
out of 100
TIME USED
03:30
of 25:00
ATTEMPTED
5/5
questions
PER QUESTION BREAKDOWN
Q1: Calculate Average
0/100
FAIL
Could not evaluate.
Time: N/A Space: N/A
Q2: Filter Positive Values
0/100
FAIL
Could not evaluate.

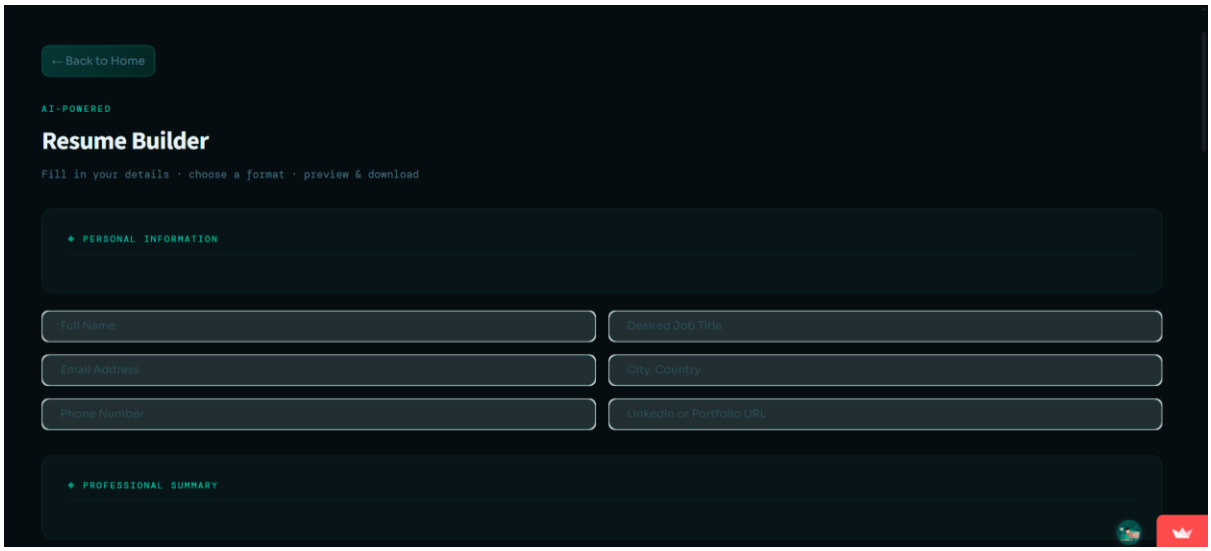
HR Interview:

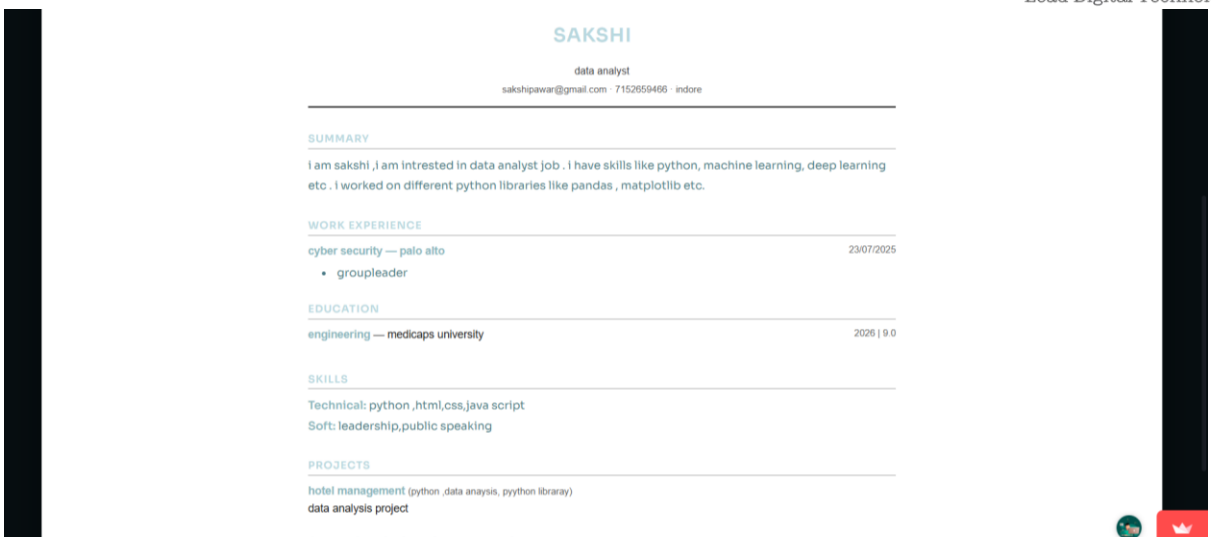






Resume Generator :





SAKSHI

data analyst

sakshipawar@gmail.com · 7152659466 · indore

SUMMARY

i am sakshi ,i am intrested in data analyst job . i have skills like python, machine learning, deep learning etc . i worked on different python libraries like pandas , matplotlib etc.

WORK EXPERIENCE

cyber security — palo alto 23/07/2025

- groupleader

EDUCATION

engineering — medicaps university 2026 | 9.0

SKILLS

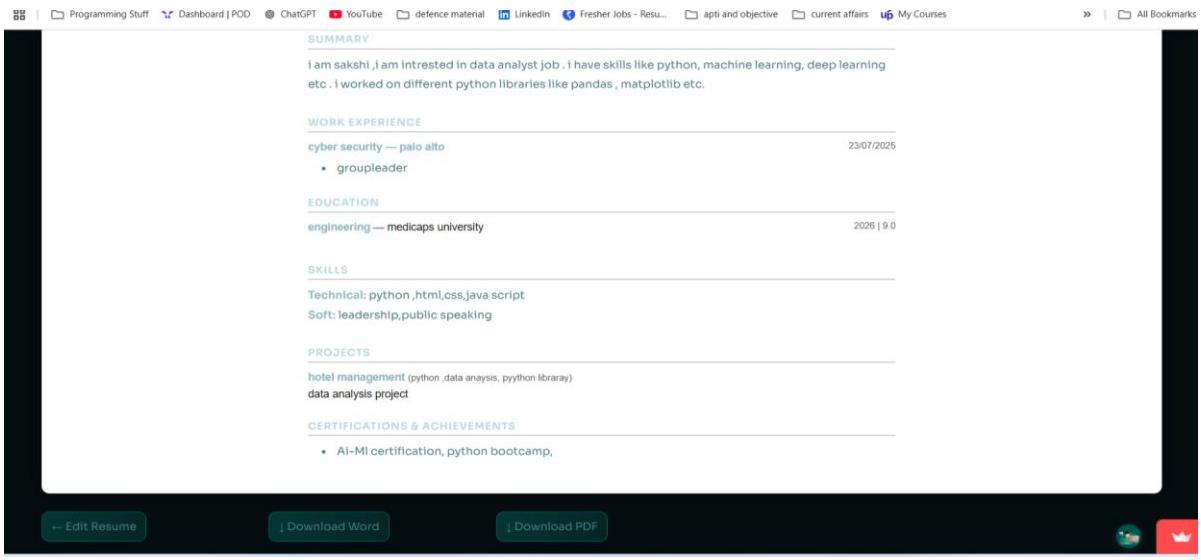
Technical: python ,html,css,java script

Soft: leadership,public speaking

PROJECTS

hotel management (python ,data anaysis, pyython libraray)

data analysis project



Programming Stuff | Dashboard | POD | ChatGPT | YouTube | defence material | LinkedIn | Fresher Jobs - Resu... | apti and objective | current affairs | My Courses | All Bookmarks

SUMMARY

i am sakshi ,i am intrested in data analyst job . i have skills like python, machine learning, deep learning etc . i worked on different python libraries like pandas , matplotlib etc.

WORK EXPERIENCE

cyber security — palo alto 23/07/2025

- groupleader

EDUCATION

engineering — medicaps university 2026 | 9.0

SKILLS

Technical: python ,html,css,java script

Soft: leadership,public speaking

PROJECTS

hotel management (python ,data anaysis, pyython libraray)

data analysis project

CERTIFICATIONS & ACHIEVEMENTS

- Ai-Ml certification, python bootcamp,

← Edit Resume | ↓ Download Word | ↓ Download PDF

2. Key Outcomes

- System generates **role-based interview questions** (e.g., Python, Data Analyst)
- Provides **real-time evaluation** of user answers
- Gives **scores with detailed feedback**
- Identifies **strengths and weaknesses** of the user
- Suggests **areas for improvement**
- Generates a **final structured PDF report**
- Ensures a **realistic interview simulation experience**
- Stores **session data for performance tracking**

Conclusion

- The project successfully implements a **multi-agent AI system** for interview preparation
- Use of agents improves **modularity and efficiency**
- Integration of LLM API enables **intelligent evaluation and feedback**
- The system provides a **personalized learning experience**
- It reduces dependency on manual interview practice
- Overall, the project is **scalable, practical, and industry-relevant**

Future Scope & Enhancements

- Add **voice-based interviews** (speech-to-text + AI evaluation)
- Include **video interview analysis** (body language, confidence)
- Expand roles (HR, Finance, Marketing, etc.)
- Add **real-time coding test environment**
- Implement **adaptive difficulty level** (easy → hard)
- Use **database for long-term performance tracking**
- Add **dashboard with analytics & progress graphs**
- Deploy as **mobile app or web platform**
- Integrate **resume analysis + personalized questions**
- Support **multi-language interviews**